## №1.16

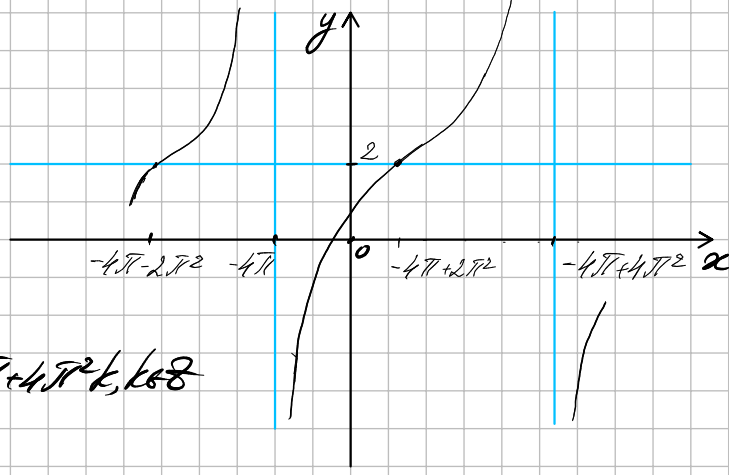
а) $y = 2 - 3\operatorname{ctg}\left(1 + \frac{x}{4\pi}\right) = f(x)$; $\lim\limits_{x \to -4\pi+} f(x) = -\infty$

$f(0) = 2 - 3\operatorname{ctg} 1 > 0$; $\lim\limits_{x \to -4\pi-} f(x) = +\infty$

$\operatorname{ctg}\left(1 + \frac{x}{4\pi}\right) = 0 \Rightarrow x = -4\pi + 2\pi^2 + \underbrace{4\pi^2 k}_{\text{период}}, k \in \mathbb{Z} \Rightarrow$

$\Rightarrow$ Пересечение $y = 2$ в точках $-4\pi + 2\pi + 4\pi^2 k, k \in \mathbb{Z}$

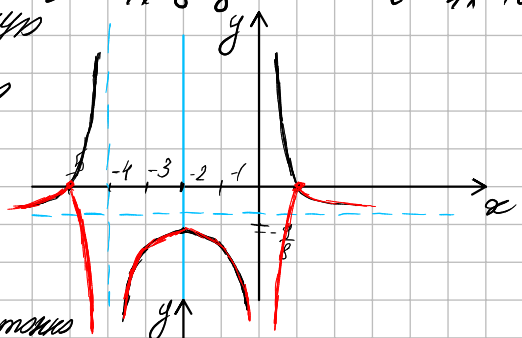
асимптоты $x = -4\pi + 4\pi^2 k, k \in \mathbb{Z}$




б) $y = -\frac{|9 - |3x+6||}{|4x+8| - 8} = -\left|\frac{9 - |3x+6|}{|4x+8| - 8}\right|$; $y' = \frac{9 - |3x+6|}{|4x+8| - 8} = \begin{cases} \frac{9-3x-6}{4x+8-8}, x \geq -2 \\ \frac{9+3x+6}{-4x-8-8}, x < -2 \end{cases} = \begin{cases} \frac{-3x+3}{4x}, x \geq -2 \\ \frac{3x+15}{-4x-16}, x < -2 \end{cases} =$

$= \begin{cases} -\frac{3}{4} + \frac{3}{4x}, x \geq -2 \\ -\frac{3}{4} - \frac{3}{4(x+4)}, x < -2 \end{cases}$; отрицательная положительная часть $y'$ относительно

$y = 0 \Rightarrow$ получим $y(-4)$




в) $y = 5^{-\frac{1}{\sqrt{x}}} = f(x)$; $\lim\limits_{x \to +\infty} f(x) = 1$

$x > 0$

$\lim\limits_{x \to 0+} f(x) = 0$; $f'(x) = \frac{\ln 5}{2x\sqrt{x}} 5^{-\frac{1}{\sqrt{x}}} > 0 \Rightarrow$ ф-я монотонно возрастает.

$\forall x \in \mathbb{R}, x > 0$,




## №2.16

$\forall \varepsilon > 0 \; \exists \delta: 0 < \left|x + \frac{1}{2}\right| < \delta \; |f(x) + 27| < \varepsilon$

$f(x) = \frac{6x^2 - 75x - 39}{3x + \frac{3}{2}} = \frac{(x + \frac{1}{2})(6x - 78)}{3(x + \frac{1}{2})}$;

$|f(x) + 27| = \left|\frac{(x+\frac{1}{2})(6x-78) + 81(x+\frac{1}{2})}{3(x+\frac{1}{2})}\right| = \left|\frac{(x+\frac{1}{2})(6x+3)}{3(x+\frac{1}{2})}\right| = \left|\frac{2}{3}(x+\frac{1}{2})^2\right| < 2\delta = \varepsilon$

$\delta = \frac{\varepsilon}{2}$

## №3.16

$\lim\limits_{x \to +0} \operatorname{tg}\frac{1}{\sqrt{x}}$; $= f(x)$ $\forall \{x_n\} \; (\forall n: x_n \neq x_0) \wedge (\lim\limits_{n \to \infty} x_n = x_0) \; \exists \lim\limits_{n \to \infty} f(x_n) = A$

$\exists \; x_n = \frac{1}{(\pi n)^2}$, $x'_n = \frac{1}{(\frac{\pi}{4} + \pi n)^2}$; $x_n \underset{n \to \infty}{\to} 0$, $x_n \neq 0$, т.к. $(\pi n)^2 > 0$

$x'_n \underset{n \to \infty}{\to} 0$, $x'_n \neq 0$, т.к. $\left(\frac{\pi}{4} + \pi n\right)^2 > 0$

$f(x_n) = \operatorname{tg}(\pi n) = 0$; $f(x'_n) = \operatorname{tg}\left(\frac{\pi}{4} + \pi n\right) = 1$, т.е.

$f(x_n) \neq f(x'_n) \Rightarrow$ противоречие, а значит, предела функции в $x_0 = 0$ $\nexists$

## №4.16

1) $\lim\limits_{x \to -1} \frac{x^2 + 3x + 2}{x^3 + 2x^2 - x - 2} = \lim\limits_{x \to -1} \frac{(x+2)(x+1)}{(x-1)(x+1)(x+2)} = -\frac{1}{2}$

2) $\lim\limits_{x \to -2} \frac{\sqrt[3]{x-6} + 2}{x+2} = \left|t = x + 2\right| = \lim\limits_{t \to 0} \frac{\sqrt[3]{t-8} + 2}{t} = \lim\limits_{t \to 0} \frac{t}{t\left(\sqrt[3]{(t-8)^2} - 2\sqrt[3]{t-8} + 4\right)} = \frac{1}{12}$

3) $\lim\limits_{x \to 2\pi} \frac{\sin 7x - \sin 3x}{e^{x^2} - e^{4\pi^2}} = \left|t = x - 2\pi\right| = \lim\limits_{t \to 0} \frac{\sin 7t - \sin 3t}{e^{t^2 + 4\pi t + 4\pi^2} - e^{4\pi^2}} = \lim\limits_{t \to 0} \frac{2\sin 2t \cos 5t}{e^{4\pi^2}(e^{t^2 + 4\pi t} - 1)} =$

$= \lim\limits_{t \to 0} \frac{4t}{e^{4\pi^2} \cdot t \cdot (t + 4\pi)} = \lim\limits_{t \to 0} \frac{4}{e^{4\pi^2}(t + 4\pi)} = \frac{1}{\pi e^{4\pi^2}}$

4) $\lim\limits_{x\to -3} \frac{\sin\left(e^{\frac{\sqrt[3]{1-x^2}}{2}} - e^{\sqrt[3]{x+2}}\right)}{\operatorname{arctg}(x+3)} = \left| t = x+3 \right| = \lim\limits_{t\to 0} \frac{\sin\left(e^{\frac{\sqrt[3]{1-t^2+6t-9}}{2}} - e^{\sqrt[3]{t-1}}\right)}{t} \to 0$

$= \lim\limits_{t\to 0} \frac{e^{\frac{\sqrt[3]{-t^2+6t-8}}{2}} - e^{\sqrt[3]{t-1}}}{t} = \lim\limits_{t\to 0} \frac{-1}{2et}\left(\sqrt[3]{t^2-6t+8} + 2\sqrt[3]{t-1}\right) = \lim\limits_{t\to 0} -\frac{1\cdot(t+2)}{2e\left(\sqrt[3]{(t-2)^2(t-4)^2} - 2\sqrt[3]{(t-1)(t-2)(t-4)} + 4\sqrt[3]{t-1}\right)} =$

$= \lim\limits_{t\to 0} -\frac{1}{e\cdot(4+4+4)} = -\frac{1}{12e}$

5) $\lim\limits_{x\to 0} (\cos 2x)^{\frac{1}{\ln(1+\sin^2 x)}} = \lim\limits_{x\to 0} \cos x^{\frac{1}{\sin^2 x}} = \lim\limits_{x\to 0} e^{\frac{\ln(\cos x)}{\sin^2 x}} = \lim\limits_{x\to 0} e^{\frac{\cos x - 1}{x^2}} =$

$= \lim\limits_{x\to 0} e^{-\frac{1}{2}\frac{x^2}{x^2}} = e^{-\frac{1}{2}}$

6) $\lim\limits_{x\to 1\pm 0} (e^{x-1} - 0{,}5)^{\frac{3x-1}{x-1}}$; $\lim\limits_{x\to 1+0} (e^{x-1} - 0{,}5)^{3+\frac{2}{x-1}} = 0$, т.к. $\lim\limits_{x\to 1+0} (e^{x-1} - 0{,}5) = \frac{1}{2} < 1$,

$\lim\limits_{x\to 1-0} (e^{x-1} - 0{,}5)^{3+\frac{2}{x-1}} = +\infty$, т.к. $\lim\limits_{x\to 1+0} \left(3 + \frac{2}{x-1}\right) = -\infty$ $\Big\}$ $\lim\limits_{x\to 1+0} \left(3 + \frac{2}{x-1}\right) = +\infty$

7) $\lim\limits_{x\to 0} \frac{\cos(1+x)}{(2+\sin\frac{1}{x})\ln(1+x)+2} \ominus \frac{\cos 1}{2}$

$\sin\frac{1}{x} \le 1 \quad \forall x \in \mathbb{R}\setminus\{0\}$

$\ln(1+x) \underset{x\to 0}{\to} 0$ $\Big| \Rightarrow \left(\left(2+\sin\frac{1}{x}\right)\ln(1+x)+2\right) \underset{x\to 0}{\to} 2$

$\cos(1+x) \underset{x\to 0}{\to} \cos 1$

8) $\lim\limits_{x\to 0} \frac{(a^x - b^x)^2}{a^{x^2} - b^{x^2}}$, $a>0$, $b>0$, $a\ne b$

$\lim\limits_{x\to 0} \frac{(a^x - b^x)^2}{a^{x^2} - b^{x^2}} = \lim\limits_{x\to 0} \frac{b^{x^2}\left(\left(\frac{a}{b}\right)^x - 1\right)^2}{b^{x^2}\left(\left(\frac{a}{b}\right)^{x^2} - 1\right)} = \lim\limits_{x\to 0} \frac{x^2 \ln^2\frac{a}{b}}{x^2 \ln\frac{a}{b}} = \ln\frac{a}{b}$.

**№ 5.16** а) $x = \frac{2t-t^2}{t-1} = \frac{-1+2t-t^2}{t-1} + \frac{1}{t-1} = \frac{1}{t-1} - (t-1)$; $y = \frac{t^2}{t-1}$

1) $\lim\limits_{t\to+\infty}\left(\frac{1}{t-1} - (t-1)\right) = -\infty$; $\lim\limits_{t\to-\infty}\left(\frac{1}{t-1} - (t-1)\right) = +\infty$; $\lim\limits_{t\to 1+0}\left(\frac{1}{t-1} - (t-1)\right) = +\infty$; $\lim\limits_{t\to 1-0}\left(\frac{1}{t-1} - (t-1)\right) = -\infty$

$\lim\limits_{t\to+\infty} \frac{t^2}{t-1} = +\infty$; $\lim\limits_{t\to-\infty} \frac{t^2}{t-1} = -\infty$; $\lim\limits_{t\to 1+0} \frac{t^2}{t-1} = \lim\limits_{t\to 1+0}(t+1) + \frac{1}{t-1} = +\infty$; $\lim\limits_{t\to 1-0}\left(t+1+\frac{1}{t-1}\right) = -\infty$

2) $\lim\limits_{t\to+\infty} \frac{y(t)}{x(t)} = \lim\limits_{t\to+\infty} \frac{t^2}{2t-t^2} = \lim\limits_{t\to+\infty} \frac{t}{2-t} = \lim\limits_{t\to+\infty} \frac{1}{\frac{2}{t}-1} = -1 = k$; $\lim\limits_{t\to+\infty}(y(t) - kx(t)) =$

$= \lim\limits_{t\to+\infty}\left(\frac{t^2}{t-1} + \frac{2t-t^2}{t-1}\right) = \lim\limits_{t\to+\infty}\left(\frac{2t}{t-1}\right) = 2 = b$ $\Big| \Rightarrow y = -x+2$ – асимптота кривой

• $\lim\limits_{t\to-\infty} \frac{y(t)}{x(t)} = -1$; $\lim\limits_{t\to-\infty}(y(t) - kx(t)) = 2 = b$ $\Big|$

• $\lim\limits_{t\to 1+0} \frac{y(t)}{x(t)} = \lim\limits_{t\to 1+0} \frac{t^2}{2t-t^2} = \lim\limits_{t\to 1+0} \frac{t}{2-t} = \lim\limits_{t\to 1-0} \frac{y(t)}{x(t)} = 1$; $\lim\limits_{t\to 1}(y(t) - kx(t)) = \lim\limits_{t\to 1}\left(\frac{t^2}{t-1} - \frac{2t-t^2}{t-1}\right) =$

$= \lim\limits_{t\to 1}\left(\frac{-2t+2t^2}{t-1}\right) = \lim\limits_{t\to 1}\left(\frac{2t(t-1)}{t-1}\right) = 2 \Rightarrow y = x+2$ – асимптота кривой.

3) $y(t) + x - 2 = \frac{t^2}{t-1} + \frac{2t-t^2}{t-1} - 2 = \frac{2t-2t+2}{t-1} > 0$ при $t>1$ – выше асимптоты $-x+2$

$< 0$ при $t<1$ – ниже асимптоты $-x+2$

$y(t) - x - 2 = \frac{t^2}{t-1} - \frac{2t-t^2}{t-1} - 2 = \frac{2t^2-2t-2t+2}{t-1} = 2(t-1) > 0$ при $t>1$ – выше $x+2$

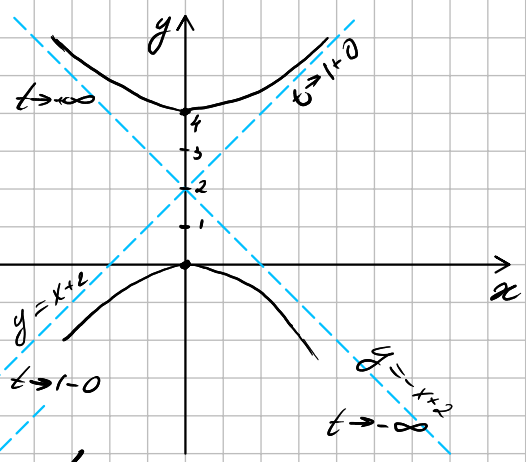
$< 0$ при $t<1$ – ниже $x+2$

4) $x'(t) = \frac{(2-2t)(t-1)-(2t-t^2)}{(t-1)^2} = \frac{-2t^2+4t-2-2t+t^2}{(t-1)^2} = \frac{-t^2+2t-2}{(t-1)^2} = -\frac{(t^2-2t+2)}{(t-1)^2} < 0 \quad \forall t \in \mathbb{R}$

$y'(t)=\frac{2t(t-1)-t^2}{(t-1)^2}=\frac{t^2-2t}{(t-1)^2}=\frac{t(t-2)}{(t-1)^2}$ (+ 0 − 1 − 2 +)

$y(0)=0,\ x(0)=0$

$y(2)=4,\ x(2)=0$



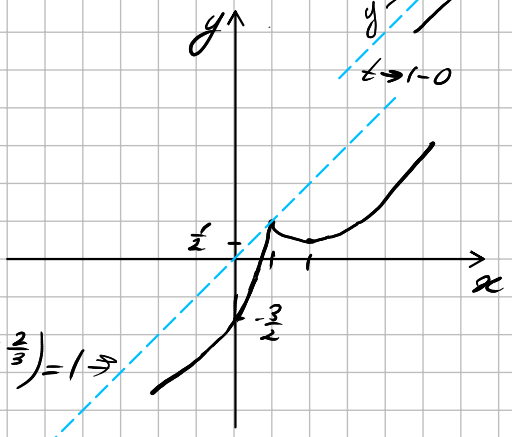

5) $y=x-\frac{3}{2}(x-1)^{\frac{2}{3}}=f(x);\ f(0)=-\frac{3}{2}.$

$y'=1-\frac{1}{\sqrt[3]{x-1}}=\frac{\sqrt[3]{x-1}-1}{\sqrt[3]{x-1}}$ (+ 1 − 2 +, 2 — min)

$f(2)=2-\frac{3}{2}=\frac{1}{2};\ f(1)=1$

$\lim\limits_{x\to 1+0} y'=-\infty;\ \lim\limits_{x\to 1-0} y'=+\infty$

$y''=\frac{1}{3\sqrt[3]{(x-1)^4}}>0$ при любых $x\neq 1$



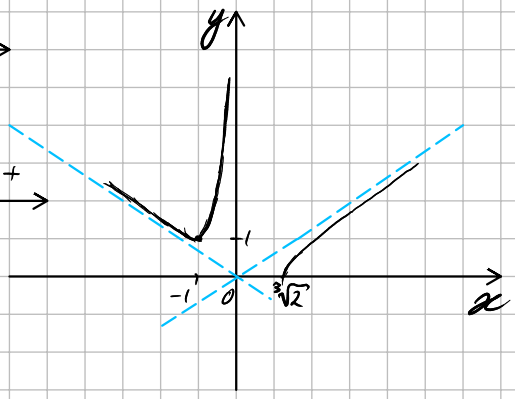

$\lim\limits_{x\to\pm\infty}\left(\frac{x-\frac{3}{2}(x-1)^{\frac{2}{3}}}{x}\right)=\lim\limits_{x\to\pm\infty}\left(1-\frac{3}{2}\left(\frac{1}{x}-\frac{1}{x^3}\right)^{\frac{2}{3}}\right)=1\Rightarrow$

$\Rightarrow y=x$ – асимптота.

6) $y=\sqrt{\frac{x^2}{3}-\frac{2}{3x}};\ \frac{x^2}{3}-\frac{2}{3x}\geq 0\Rightarrow\frac{x^3-2}{3x}\geq 0$ (+ 0 − $\sqrt[3]{2}$ +)

$x\in(-\infty;0)\cup[\sqrt[3]{2};+\infty)\Rightarrow y(\sqrt[3]{2})=0$

$y'=\left(\sqrt{\frac{x^3-2}{3x}}\right)'=\frac{9x^3-3x^3+6}{6x\sqrt{\frac{x^3-2}{3x}}}=\frac{x^3+1}{\sqrt{x(x^3-2)}}$ (− −1 + 0 + $\sqrt[3]{2}$ +, −1 — min)

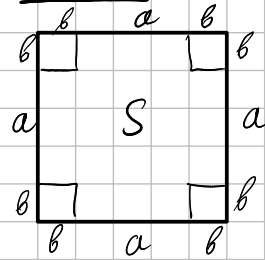



$f(-1)=1;\ \lim\limits_{x\to 0-0} y'=+\infty;\ \lim\limits_{x\to\sqrt[3]{2}+0} y'=+\infty$

$y=\sqrt{\frac{x^2}{3}-\frac{2}{3x}}=|x|\sqrt{\frac{1}{3}-\frac{2}{3x^3}}\Rightarrow\lim\limits_{x\to+\infty}y=\frac{x}{\sqrt{3}}$ – асимптота

$\Rightarrow\lim\limits_{x\to-\infty}y=-\frac{x}{\sqrt{3}}$ – асимптота

**№6.16** $V\to\max;\ S=(a+2b)^2\Rightarrow\sqrt{S}=a+2b\Rightarrow a=\sqrt{S}-2b$




$V=a^2\cdot b=(S-4b\sqrt{S}+4b^2)b\Rightarrow V'(b)=-4b\sqrt{S}+8b^2+S-4b\sqrt{S}+4b^2=$

$=12b^2-8b\sqrt{S}+S=0\Rightarrow b=\frac{4\sqrt{S}\pm\sqrt{16S-12S}}{12}=\frac{4\sqrt{S}\pm 2\sqrt{S}}{12}\Rightarrow\left[\begin{array}{l}b=\frac{\sqrt{S}}{2}\\ b=\frac{\sqrt{S}}{6}\end{array}\right.\Rightarrow$

$\Rightarrow b=\frac{\sqrt{S}}{6}$

(+ $\frac{\sqrt{S}}{6}$ max − $\frac{\sqrt{S}}{2}$ min +)

**№7.16** а) $\lim\limits_{x\to 0}\frac{\ln(\cos x+x\sin x)-\frac{x^2}{2}e^x}{\frac{x}{2}\sqrt[3]{1-x}+\sqrt{1+\frac{x^2}{3}}-\sin(\frac{x}{2})-1}$ ⊖

$\sqrt[3]{1-x}=(1-x)^{\frac{1}{3}}=1-\frac{x}{3}+\frac{\frac{1}{3}(-\frac{2}{3})}{2}x^2-\frac{(\frac{1}{3})(-\frac{2}{3})(-\frac{5}{3})}{3\cdot 2}x^3+\bar{o}(x^3)=1-\frac{x}{3}-\frac{1}{9}x^2-\frac{5}{81}x^3+\bar{o}(x^3)$

$\frac{x}{2}\cdot\sqrt[3]{1-x}=\frac{x}{2}-\frac{x^2}{6}-\frac{x^3}{18}+\bar{o}(x^3)$

$(1+\frac{x^2}{3})^{\frac{1}{2}}=1+\frac{x^2}{6}-\frac{1}{12}x^4+\bar{o}(x^4)$

$\sin(\frac{x}{2})=\frac{x}{2}-\frac{x^3}{8\cdot 3!}+\bar{o}(x^3)$

$\Rightarrow\frac{x}{2}\sqrt[3]{1-x}+\sqrt{1+\frac{x^2}{3}}-\sin(\frac{x}{2})-1=\frac{x}{2}-\frac{x^2}{6}-\frac{x^3}{18}+\bar{o}(x^3)+1+\frac{x^2}{6}-$

$-\frac{x}{2}+\frac{x^3}{8\cdot 3!}+\bar{o}(x^3)+=\frac{-8x^3+3x^3}{3\cdot 8\cdot 3!}+\bar{o}(x^3)=-\frac{5x^3}{24\cdot 3!}+\bar{o}(x^3)$

$\sin x=x-\frac{x^3}{3!}+\bar{o}(x^3)\Rightarrow x\sin x=x^2+\bar{o}(x^3)$

$\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}+\bar{o}(x^4)$

$\Rightarrow\cos x+x\sin x=1+\frac{x^2}{2}+\bar{o}(x^3)$

⊖ $\lim\limits_{x\to 0}\frac{\ln(1+\frac{x^2}{2}+\bar{o}(x^3))}{-5/(24\cdot 3!)x^3+\bar{o}(x^3)}=\lim\limits_{x\to 0}\left(\frac{x^2}{2}\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\bar{o}(x^3)\right)\right)/\left(-\frac{5x^3}{24\cdot 3!}+\bar{o}(x^3)\right)=$

$= \lim\limits_{x\to 0} \frac{\frac{x^2}{2}+\bar{o}(x^3)}{-\frac{5x^3}{24\cdot 3!}+\bar{o}(x^3)} \leftarrow \lim\limits_{x\to 0} \frac{\frac{x^2}{2}+\frac{x^3}{2}+\bar{o}(x^3)}{-\frac{5x^3}{24\cdot 3!}+\bar{o}(x^3)} = \frac{1}{2}\cdot\frac{12\cdot 24\cdot 3!}{5} = \frac{72}{5} = 14{,}4$

б) $\lim\limits_{x\to 0} (\sqrt[3]{1+3x}-\operatorname{tg}\sin x+x^2)^{\frac{1}{\operatorname{arctg} x - x\cos x}} = \lim\limits_{x\to 0} e^{\frac{1}{\operatorname{arctg} x - x\cos x}\ln(\sqrt[3]{1+3x}-\operatorname{tg}\sin x+x^2)} \ominus$

$\triangle \frac{1}{\operatorname{arctg} x - x\cos x} = \frac{1}{x-\frac{x^3}{3}+\bar{o}(x^3)-x+\frac{x^3}{2}} = \frac{1}{\frac{x^3}{6}+\bar{o}(x^3)}$

$\operatorname{arctg} x = x - \frac{x^3}{3} + \bar{o}(x^3)$

$x\cos x = x - \frac{x^3}{2!} + \bar{o}(x^3)$

$\triangle \sqrt[3]{1+3x} = (1+3x)^{\frac{1}{3}} = 1+x-2x^2+10x^3+\bar{o}(x^3) \Rightarrow \sqrt[3]{1+3x}+x^2 = 1+x-x^2+10x^3+\bar{o}(x^3)$

$\triangle \operatorname{tg}\sin x = x + \frac{x^3}{3!} + \bar{o}(x^3)$

$\operatorname{tg}\sin 0 = 0;\ (\operatorname{tg}\sin x)' = \frac{\cos x}{\cos^2 \sin x} \Rightarrow (\operatorname{tg}\sin x)'(0) = 1;\ (\operatorname{tg}\sin x)'' = \left(\frac{\cos x}{\cos^2 \sin x}\right)' =$

$= \frac{-\sin x \cos^2 \sin x - \cos^2 x \sin 2\sin x}{\cos^4 \sin x} \Rightarrow (\operatorname{tg}\sin x)''(0) = 0;$

$(\operatorname{tg}\sin x)''(0) = 1 \Rightarrow \lim\limits_{x\to 0} \ln(\sqrt[3]{1+3x}-\operatorname{tg}\sin x+x^2) = \lim\limits_{x\to 0} \ln(1+x-x^2+10x^3+x+\frac{x^3}{3!}+\bar{o}(x^3))$

$= \lim\limits_{x\to 0} (2x - x^2 + \frac{61}{6}x^3 + \bar{o}(x^3));$

$\ominus \lim\limits_{x\to 0} e^{\frac{6}{x^3+\bar{o}(x^3)}\cdot\frac{(12x-6x^2+61x^3+\bar{o}(x^3))}{6}} = \lim\limits_{x\to 0} e^{\frac{1}{x^3}\cdot x^3(\frac{12}{x^2}-\frac{6}{x}+61)} = \lim\limits_{x\to 0} e^{\frac{12-6x}{x^2}+61} =$

$= +\infty$, т.к. $\frac{12-6x}{x^2} \xrightarrow[x\to 0]{} +\infty$, $e > 1$

**№ 8.16** $f(x) = 2x + \frac{1}{1-3x}$. Равномерная непрерывность: $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, x'\ |x'-x| < \delta \Rightarrow |f(x')-f(x)| < \varepsilon$

а) $X = [1; +\infty)$

$|f(x')-f(x)| = |2x' + \frac{1}{1-3x'} - 2x - \frac{1}{1-3x}| \le |2x'-2x| + |\frac{1}{1-3x'} - \frac{1}{1-3x}| =$

$= 2|x'-x| + |\frac{-3x+3x'}{(1-3x)(1-3x')}| < 2|x'-x| + \frac{3}{4}|x'-x| = 3|x'-x| < \varepsilon \quad \delta = \frac{\varepsilon}{3}$

б) $X = (\frac{1}{3}; 1)$ $\quad x' = \frac{1}{3} + \frac{1}{4n}$ $\quad x = \frac{1}{3} + \frac{1}{2n}$ $\quad |f(x')-f(x)| = |\frac{2}{3} + \frac{1}{2n} - \frac{2}{3} - \frac{1}{n} + \frac{1}{1+\frac{3}{4n}} - \frac{1}{1+\frac{3}{2n}}| \ominus$

$\ominus |-\frac{1}{2n} + \frac{4}{3}n - \frac{2}{3}n| = |\frac{2}{3}n - \frac{1}{2n}| > |\frac{n^2-1}{2n}| = |\frac{n-\frac{1}{n}}{2}| \xrightarrow[n\to\infty]{} \infty$

$|x'-x| = |\frac{1}{4n} - \frac{1}{2n}| = \frac{1}{4n} \xrightarrow[n\to\infty]{} 0$

$\Rightarrow$ ф-я не равномерно непрерывна на $(\frac{1}{3}; 1)$